

Indian Medicine Landscape

Uncovering market structure, price patterns & formulation science

A-Z Medicine Dataset of India • 253,973 raw records

Python • Pandas • NumPy • Plotly

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Why does this dataset matter?

The analytical problem

A catalogue with more than 250,000 medicine records is too large to understand by looking at individual rows. We need a structured way to convert those records into patterns that can be explained clearly.

Our approach

- Clean the raw records
- Measure important catalogue characteristics
- Compare groups and distributions
- Use visuals to communicate evidence

What we want to understand

- Manufacturer representation
- Medicine price patterns
- Differences between manufacturer price distributions
- Frequently listed active ingredients
- Single vs combination formulations

1. Introduction

India is globally recognized as the "Pharmacy of the World", serving as one of the largest producers and consumers of generic and branded medications. The Indian pharmaceutical marketplace contains a diverse ecosystem of manufacturers, pricing structures, and chemical formulations spanning acute and chronic therapies..

In this project, we analyze a comprehensive dataset containing over 250,000 medicine records from across India to investigate fundamental questions :

Catalogue Concentration : How concentrated are medicine listings among industry leaders versus smaller regional producers?

Pricing & Affordability: How are medicine prices distributed across the country, and what share of treatments are accessible at low cost?

Manufacturer Price Distributions: Do leading pharmaceutical manufacturers exhibit uniform pricing, or do their catalogue spreads reflect distinct price distributions?

Formulated Chemical Ingredients: Which active pharmaceutical ingredients (APIs) are most frequently represented across the catalogue?

Formulation Architecture: How prevalent are combination formulations compared to single-ingredient medications?

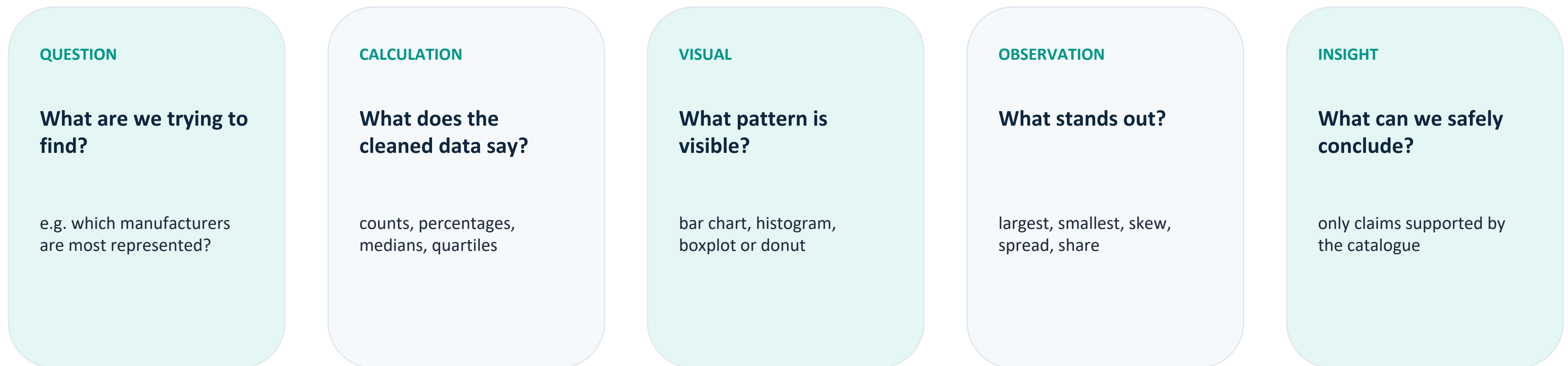
The Data Story Philosophy

Rather than jumping straight to modeling, this project follows an evidence-based storytelling methodology:

Raw Ingestion (253,973)→Data Governance \& Scope Validation→Clean Analytical Dataset (243,803)→5 Core Insights→Interactive Plotly Visuals→Synthesis

How the data becomes a story

We do not decide the findings first. Each insight follows a simple evidence chain:



We started with the raw medicine dataset and followed a simple analytical process: clean the data, calculate relevant statistics, visualise the patterns, and interpret the findings. Each insight is based on what the cleaned data actually shows.

This keeps the presentation data-driven: the graphs are not decoration; they are the evidence used to support each insight.

Research questions guiding the analysis

Objective

The objective of this project is to transform a large Indian medicine catalogue into a clear, evidence-based data story using Python and data visualisation. Starting with 253,973 raw medicine records, we aim to clean and validate the dataset, explore its important characteristics, identify meaningful patterns, and communicate those findings through charts and statistical evidence. The analysis focuses on understanding how medicines are distributed across manufacturers, how prices are distributed, how pricing patterns vary between major manufacturers, which active ingredients appear most frequently, and how common combination formulations are. The overall goal is not simply to present numbers, but to use the data to explain the structure and patterns visible within the medicine catalogue in a simple and understandable way.

Scope

- Full cleaned dataset used
- No preselected manufacturers
- No 5,000-cheapest subset

Five research questions

- Who is most represented among manufacturers?
- What does the overall price distribution look like?
- How do major manufacturers differ in price distribution?
- Which active ingredients are most frequently listed?
- How common are combination formulations?

Dataset overview & quality audit

253,973

raw records

9

attributes

7,611

manufacturers

7,750

primary active ingredients

Variables used in the story

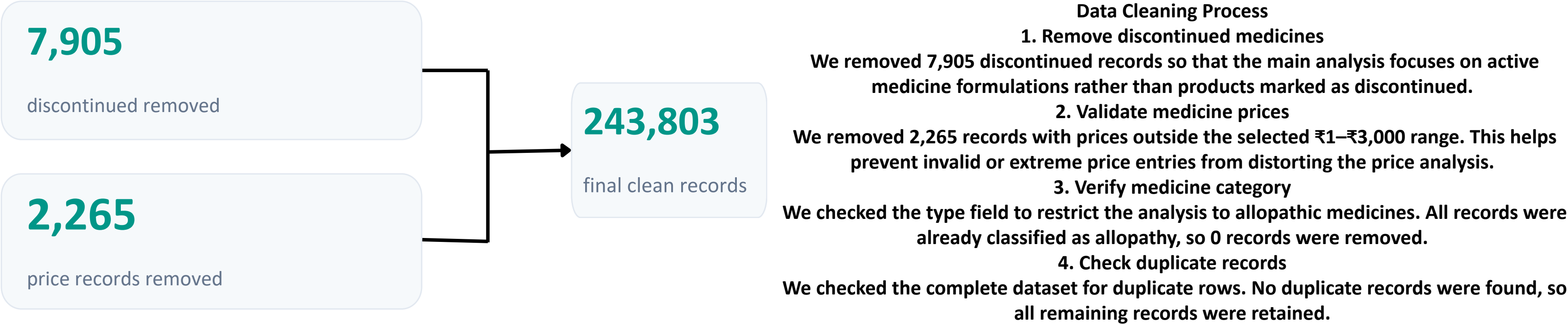
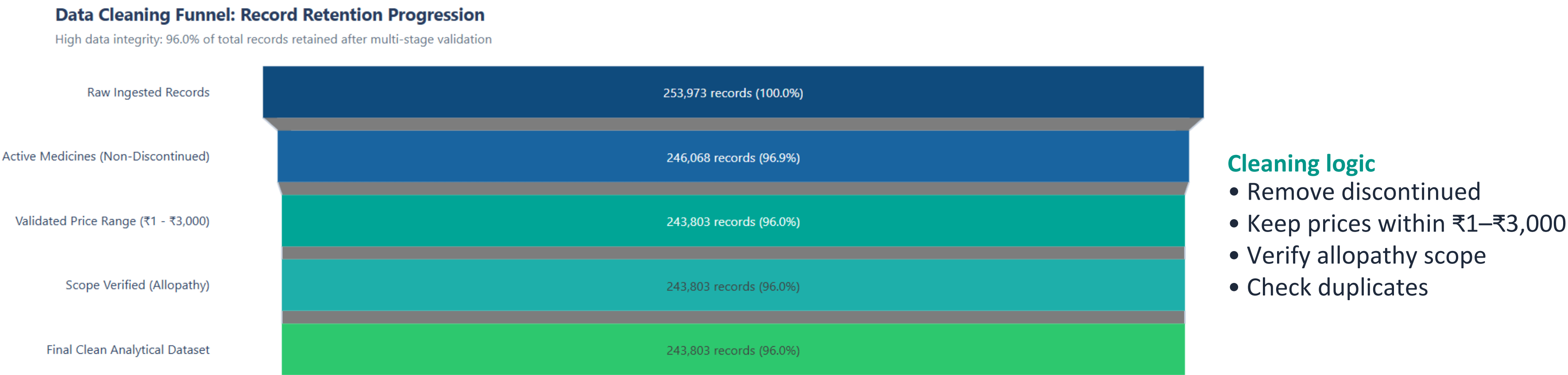
- name — medicine / brand
- price(₹) — maximum retail price
- manufacturer_name — producer
- short_composition1 — primary ingredient
- short_composition2 — secondary ingredient
- Is_discontinued / type — status and scope

Initial audit findings

- Primary fields contain complete values
- 7,905 records are discontinued
- All source records are classified as allopathy
- Missing secondary composition is structurally meaningful for single-ingredient records

Our analysis is based on the A-Z Medicine Dataset of India, which contains 253,973 medicine records across 9 attributes. The dataset includes important fields such as the medicine or brand name, maximum retail price, manufacturer name, primary and secondary active ingredients, discontinued status, and medicine type. These variables allow us to study four major dimensions of the catalogue: manufacturer representation, price distribution, active ingredient frequency, and formulation structure. Before beginning the main analysis, we conducted an initial data-quality audit by examining the dataset structure, data types, missing values, duplicate records, and discontinued medicines. The audit identified 7,905 discontinued records, while all records were already classified as allopathy. The dataset contains 7,611 manufacturers and 7,750 distinct primary active ingredients, providing a broad base for exploring patterns across the Indian medicine catalogue.

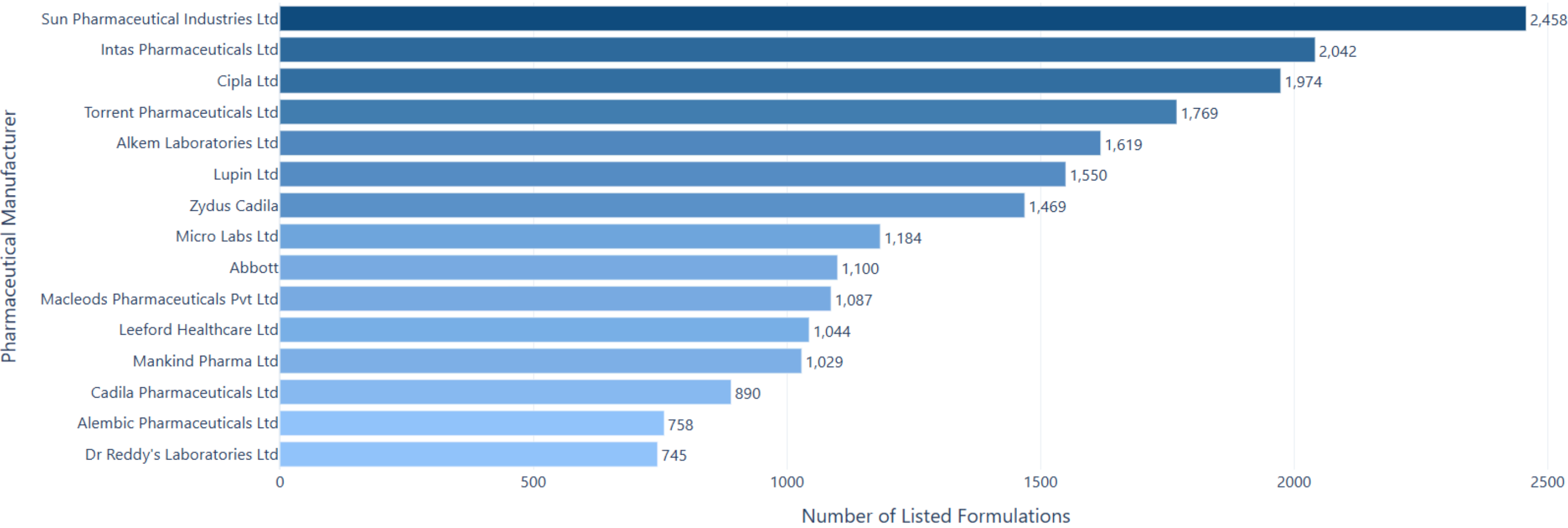
Cleaning pipeline: from raw data to analytical baseline



Insight 1 — A fragmented manufacturer landscape

Insight 1: Top 15 Pharmaceutical Manufacturers by Catalogue Volume

Sun Pharma (2,458), Intas (2,042), and Cipla (1,974) lead in product listing count



What are we measuring?

Number of medicine formulations listed for each manufacturer. This is catalogue representation, not sales or revenue market share.

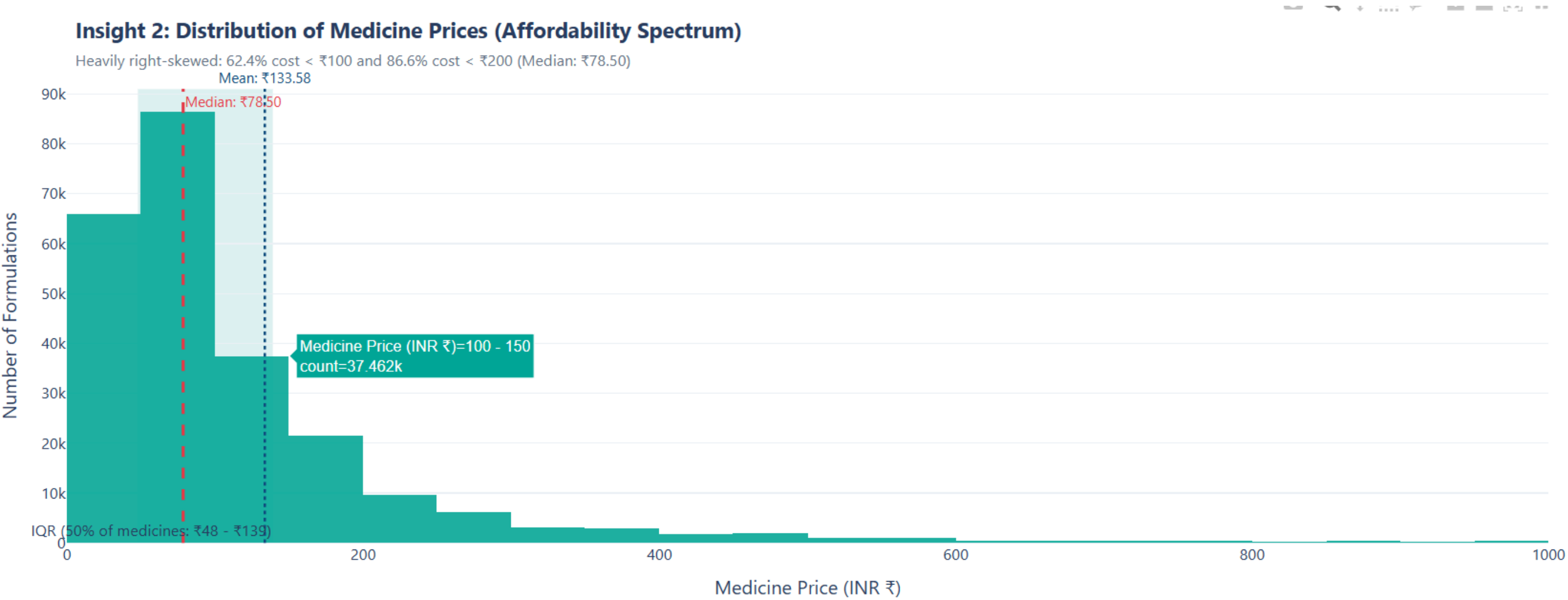
Evidence

- Sun Pharma — 2,458
- Intas — 2,042
- Cipla — 1,974
- Top 10 — 6.67% of listings

Meaning

The manufacturer analysis shows that Sun Pharma leads the catalogue with 2,458 formulations, followed by Intas with 2,042 and Cipla with 1,974. However, the top 10 manufacturers together represent only 6.67% of all 243,803 listings, while the top 50 account for 15.35%. This means that the catalogue is not concentrated among a small number of leading companies. Instead, more than 7,500 other manufacturers collectively contribute 84.65% of the listings, revealing a strong long-tail structure in the dataset.

Insight 2 — Price distribution is strongly right-skewed



What are we measuring?

The histogram shows how frequently medicine prices occur across ranges. Median and mean help describe the centre of the distribution.

Evidence

₹78.50

median

86.6%

under ₹200

₹133.58

mean

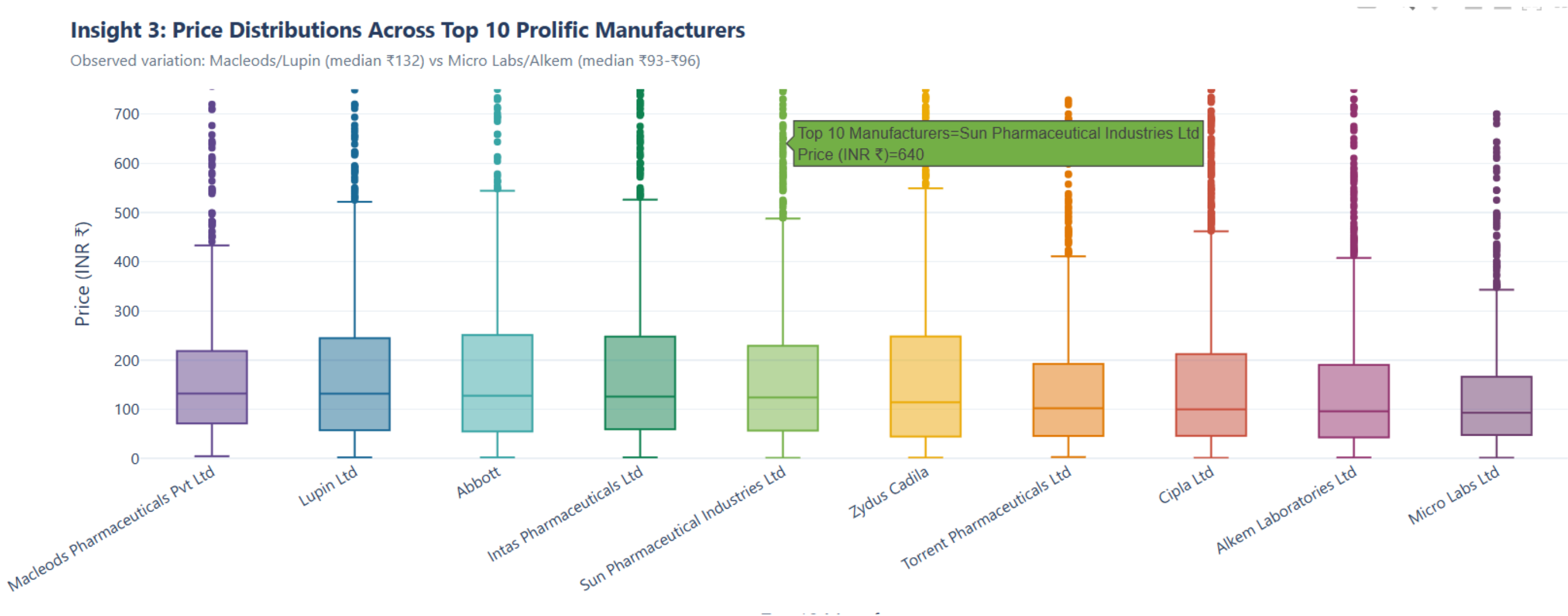
₹48—₹139

middle 50%

Meaning

The price distribution is strongly concentrated at the lower end. The median price is ₹78.50, while the mean is higher at ₹133.58 because a smaller number of expensive medicines create a long right-hand tail. The middle 50% of medicines fall between ₹48 and ₹139, and 86.6% of all listings are priced below ₹200. Overall, the dataset shows that lower-priced medicines make up the majority of the catalogue, while a relatively small group of higher-priced medicines increases the average price.

Insight 3 — Manufacturers show different price distributions



Why this visual?

A boxplot shows the median, middle 50% and spread for each manufacturer, making distributions easier to compare than averages alone.

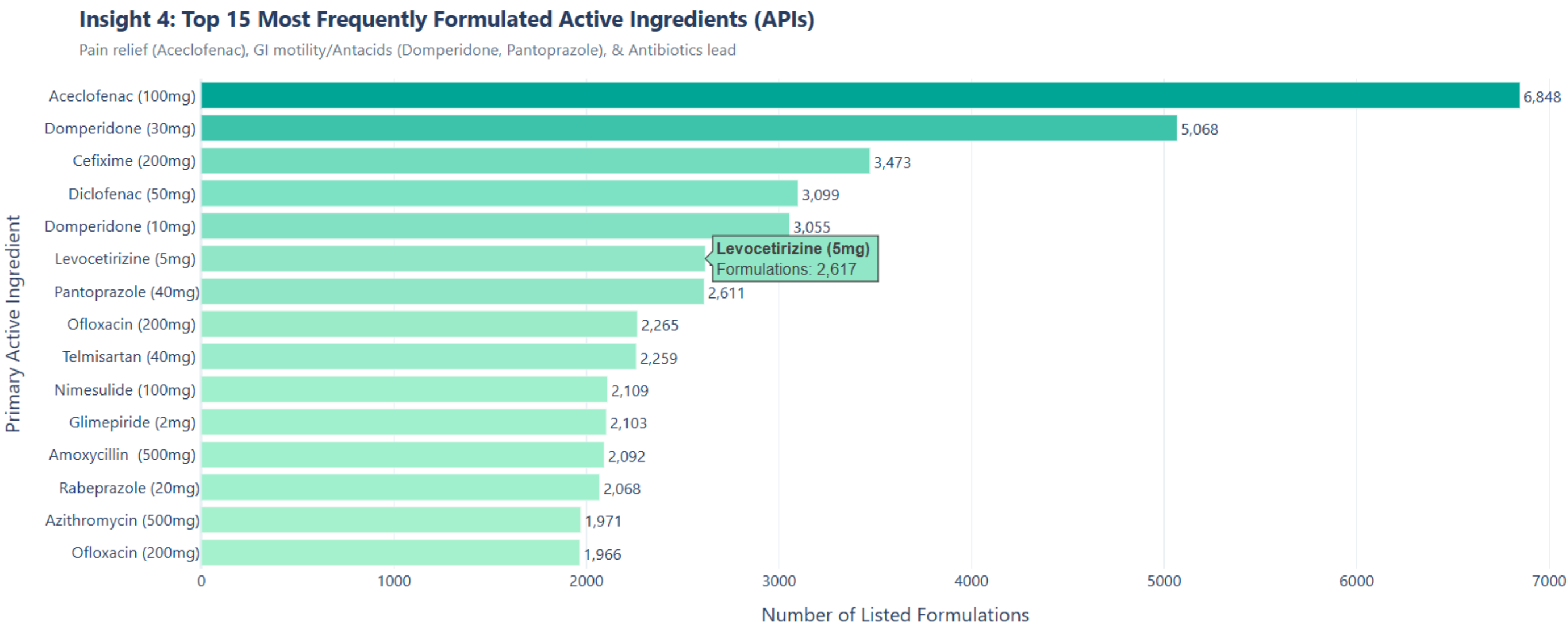
Evidence

- Macleods median — ₹132.00
- Lupin median — ₹131.75
- Micro Labs median — ₹93.00
- Zydus IQR — ₹203.10

Meaning

The boxplot shows that major manufacturers do not have the same price distribution. Macleods has the highest median at ₹132, followed closely by Lupin at ₹131.75, while Micro Labs has the lowest median at ₹93. Zydus Cadila has the widest spread, with an IQR of ₹203.10, indicating much greater variation in the prices of its listed formulations. Overall, the comparison shows clear differences in the price profiles of major manufacturers, although the dataset alone cannot determine the reasons behind these differences.

Insight 4 — A small set of active ingredients dominates listings



What are we measuring?

Frequency of a primary active ingredient across catalogue formulations. It measures product frequency, not patient prescriptions or disease prevalence.

Evidence

- Aceclofenac 100mg — 6,848
- Domperidone variants — 8,100+
- Cefixime 200mg — 3,473
- Telmisartan 40mg — 2,259

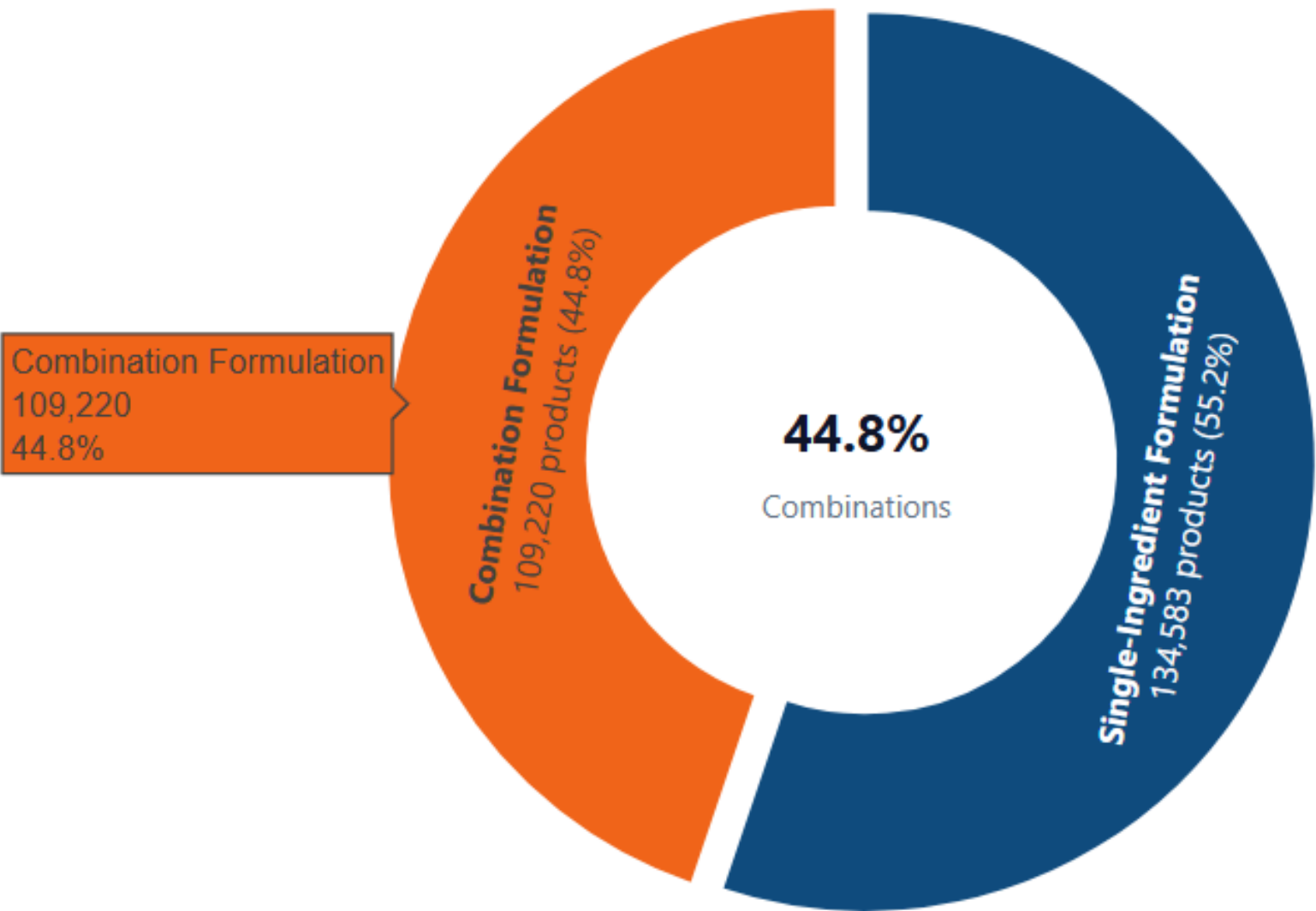
Meaning

The chart shows that a relatively small group of active ingredients appears across thousands of medicine formulations. Aceclofenac is the most frequently listed ingredient with 6,848 formulations, while Domperidone appears in more than 5,068 formulations across the two strengths shown. Other highly represented ingredients include Cefixime, Diclofenac, Pantoprazole and Telmisartan. Together, these ingredients span four broad areas pain and inflammation, gastrointestinal care, anti-infectives and chronic disease management showing that the medicine catalogue contains especially high product representation in these therapeutic areas.

Insight 5 — Combination formulations are substantial

Insight 5: Single-Ingredient vs Combination Formulations

Substantial combination presence: 44.8% of catalogue listings feature multiple active ingredients



How is the split calculated?

A record is classified as a combination formulation when a secondary active composition is present. This is a catalogue-structure measure.

55.2%

single-ingredient

44.8%

combination

Evidence & meaning

The catalogue contains a substantial mix of single-ingredient and combination formulations. Out of 243,803 clean records, 134,583 (55.2%) are single-ingredient formulations, while 109,220 (44.8%) contain a secondary active ingredient. This means combination formulations represent nearly half of the entire catalogue, showing that multi-ingredient products are an important part of the medicine landscape represented in the dataset.

The Story the Data Revealed

We started with more than 253,000 medicine records and wanted to understand what was hidden inside the catalogue.

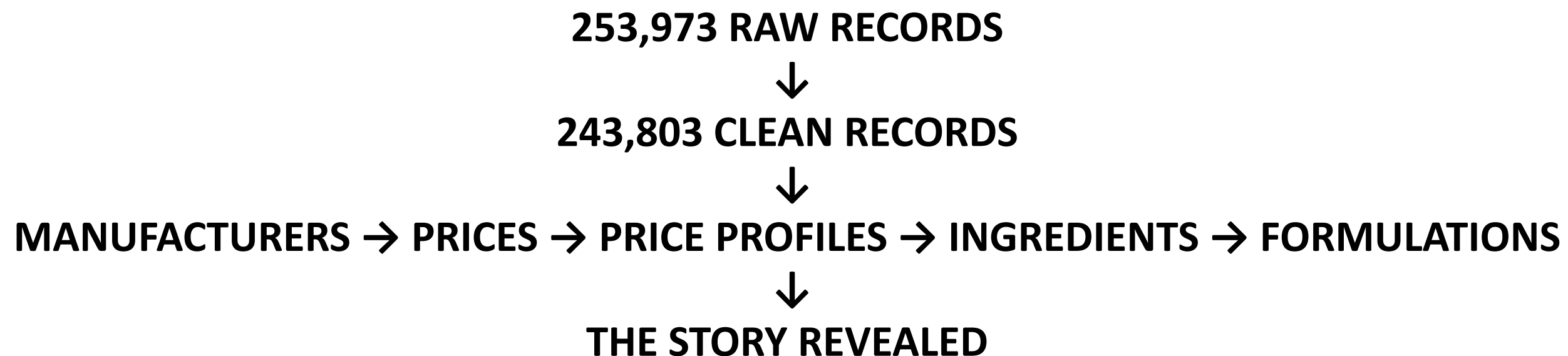
First, we found a highly distributed manufacturer landscape: although major companies lead individually, more than 7,500 other manufacturers contribute most of the remaining listings.

We then looked at price, where the median was ₹78.50 and 86.6% of listings were below ₹200.

Comparing manufacturers showed that their price distributions are not identical. Looking at composition, we found recurring active ingredients across pain, gastrointestinal care, anti-infectives and chronic disease management.

Finally, 44.8% of the catalogue consisted of combination formulations.

Together, these findings turn a huge medicine catalogue into a clear story about manufacturers, prices, ingredients and formulations.



From 253,973 records to an evidence-based data story

What we conclude

- The catalogue is highly distributed across manufacturers.
- Prices are concentrated toward the lower end with a long tail.
- Manufacturer price distributions differ.
- Recurring ingredients span several broad healthcare domains.
- Combination formulations are a substantial catalogue segment.

Limitations to remember

- Catalogue records are not patient prescriptions.
- MRP is not necessarily the final transaction price.
- Frequency does not measure disease burden or demand.
- Price patterns do not prove corporate strategy.
- Combination classification is based on secondary-composition presence.

Data → Evidence → Insight

THANK YOU